

Floor Planning/Bin Packing

Linear Program

maximise $P = 2x + 3.5y$

subject to:

$$3x + 2y \leq 14$$

$$2x + 3y \leq 12.5$$

$$-x + y \leq 1$$

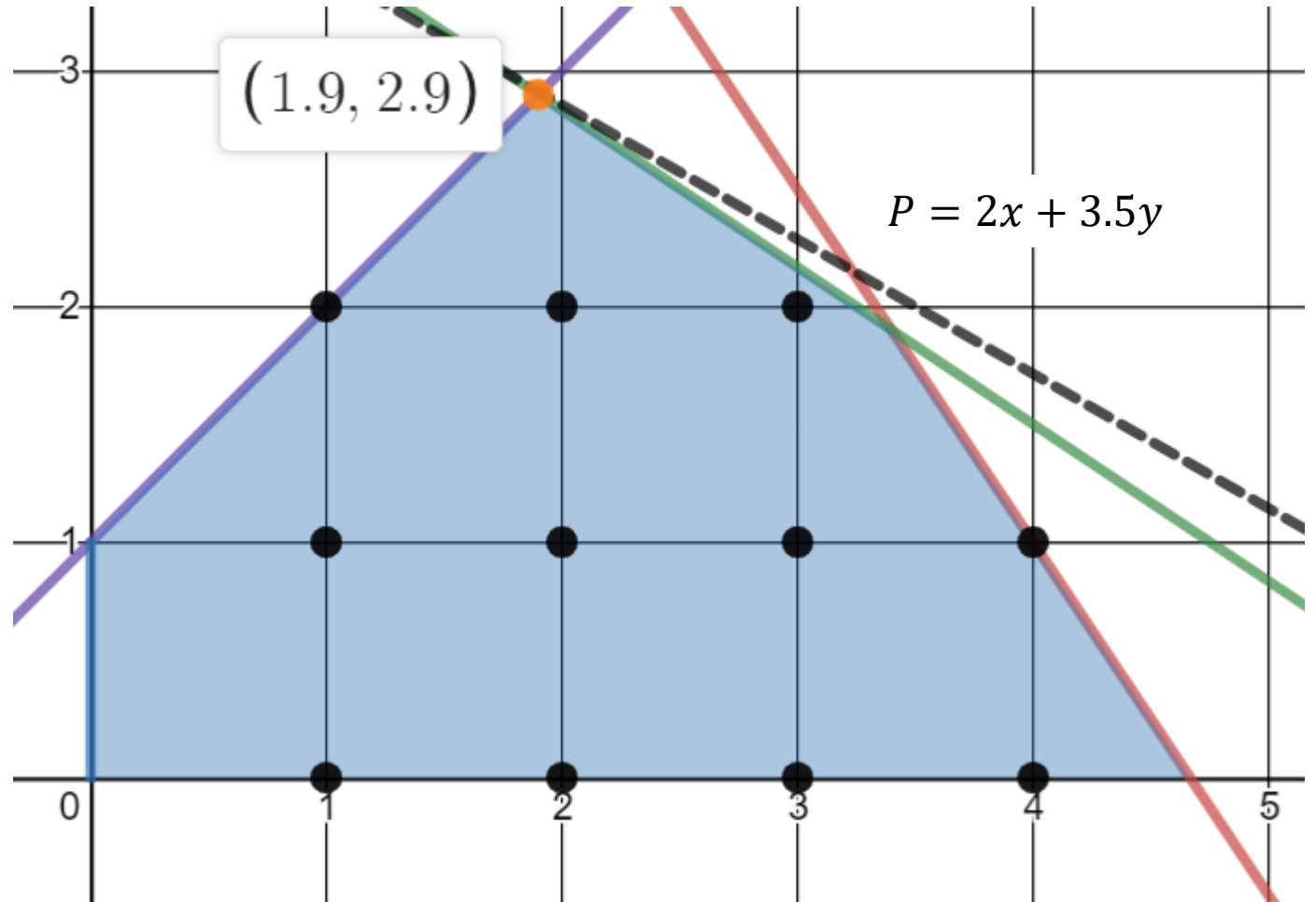
and

$$x, y \geq 0$$

Optimal Solution

$$x^* = (x, y) = (1.9, 2.9)$$

$$P^* = 13.95$$



Mixed-Integer Program

maximise $P = 2x + 3.5y$

subject to:

$$3x + 2y \leq 14$$

$$2x + 3y \leq 12.5$$

$$-x + y \leq 1$$

and

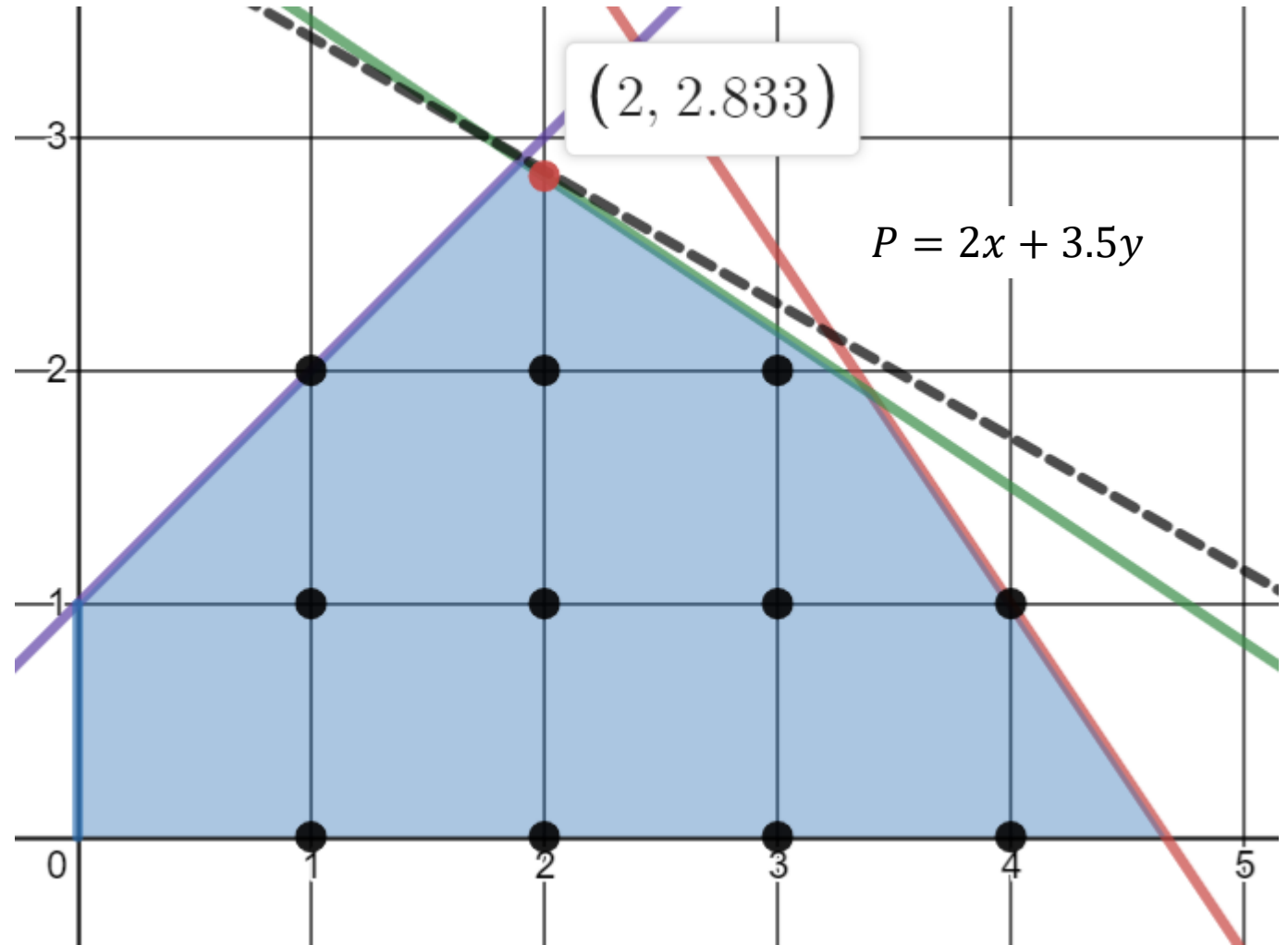
$$x, y \geq 0$$

$$x \in \mathbb{Z}$$

Optimal Solution

$$\mathbf{x}^* = (x, y) = (2, 2.833)$$

$$P^* = 13.917$$



Mixed-Integer Solution (problem based)

maximise $P = 2x + 3.5y$

subject to:

$$3x + 2y \leq 14$$

$$2x + 3y \leq 12.5$$

$$-x + y \leq 1$$

and

$$x, y \geq 0$$

$$x \in \mathbb{Z}$$

Optimal Solution

$$x^* = (x, y) = (2, 2.833)$$

$$P^* = 13.917$$

```
1 - clear all;
2 - % define integer variables
3 - x1 = optimvar('x1',1,'LowerBound',0,'UpperBound',inf,'Type','integer');
4 - x2 = optimvar('x2',1,'LowerBound',0,'UpperBound',inf);
5 - % define objective function values (row vector)
6 - f = [2 3.5];
7 - % create optimisation problem
8 - prob = optimproblem('Objective',sum(f(1)*x1+f(2)*x2),'ObjectiveSense','max');
9 - % define the inequality constraints
10 - c1 = 3*x1 + 2*x2 <= 14;
11 - c2 = 2*x1 + 3*x2 <= 12.5;
12 - c3 = -x1 + x2 <= 1;
13 - % add them to the optimisation problem
14 - prob.Constraints.c1 = c1;
15 - prob.Constraints.c2 = c2;
16 - prob.Constraints.c3 = c3;
17 -
18 - % convert the optimisation problem
19 - problem = prob2struct(prob);
20 - % solve and return the solution, function value, exitflag and output
21 - [sol,fval,exitflag,output] = intlinprog(problem)
```

sol =
2.0000
2.8333
fval =
-13.9167

2 Dimensional Floor Planning

Floor Planning

Pack the following 4 tables into the smallest area possible.

We will build a model which minimises the squared area. Even this is complex

$$t_1 - (w_1, h_1) = (4, 5)$$

$$t_2 - (w_2, h_2) = (3, 7)$$

$$t_3 - (w_3, h_3) = (6, 4)$$

$$t_5 - (w_4, h_4) = (7, 7)$$

Each t_i has co-ordinates (x_i, y_i) for it's position

Floor Planning

$$t_1 - (w_1, h_1) = (4, 5)$$

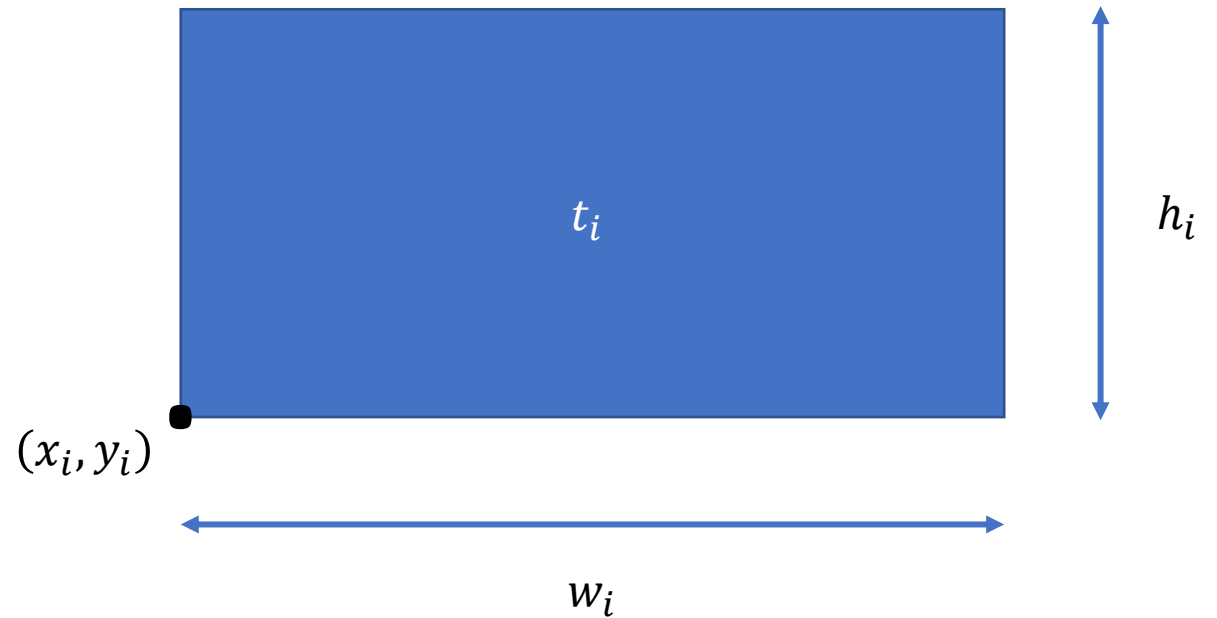
$$t_2 - (w_2, h_2) = (3, 7)$$

$$t_3 - (w_3, h_3) = (6, 4)$$

$$t_5 - (w_4, h_4) = (7, 7)$$

Each t_i has:

- co-ordinates (x_i, y_i)
- Width - w_i
- Height - h_i



Non-overlapping constraints

Make sure that the tables don't overlap.

Consider table t_1 and table t_2

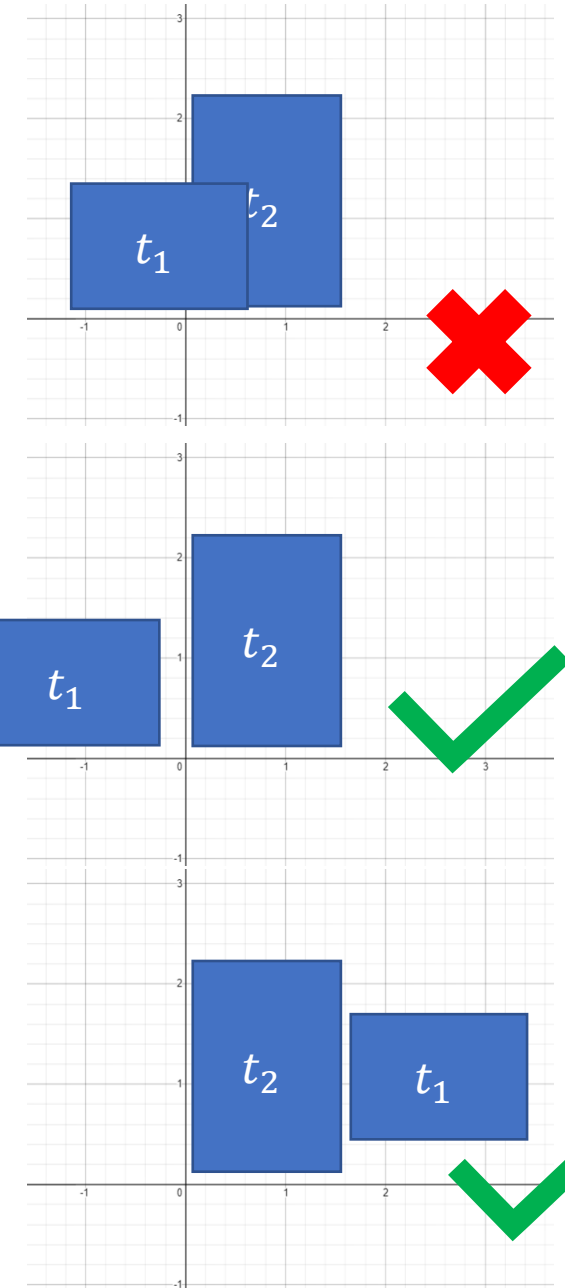
Either

$$x_1 + w_1 \leq x_2$$

Or

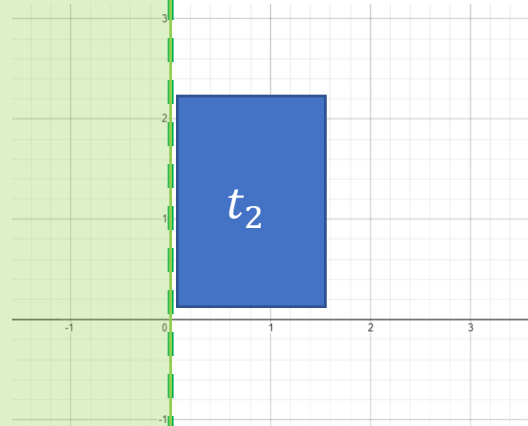
$$x_2 + w_2 \leq x_1$$

Only one can hold!



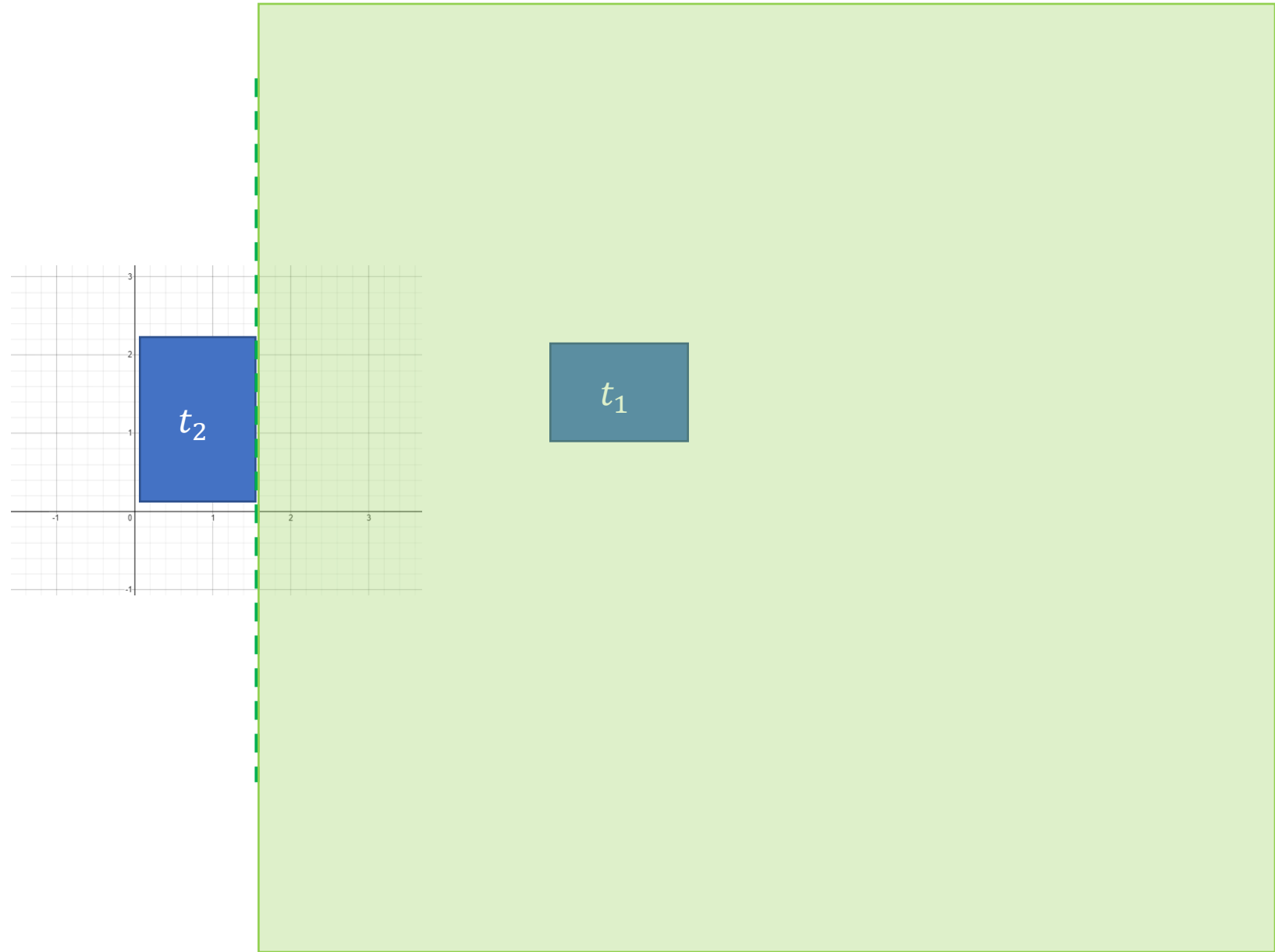
Intuition – 4 possible situations

$$x_1 + w_1 \leq x_2$$



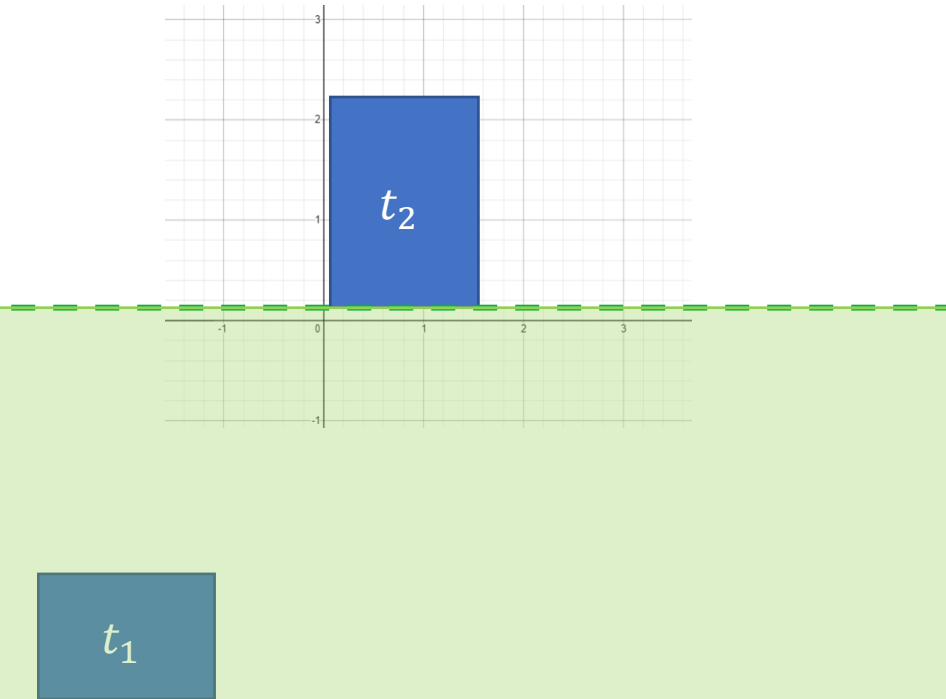
Intuition – 4 possible situations

$$x_2 + w_2 \leq x_1$$



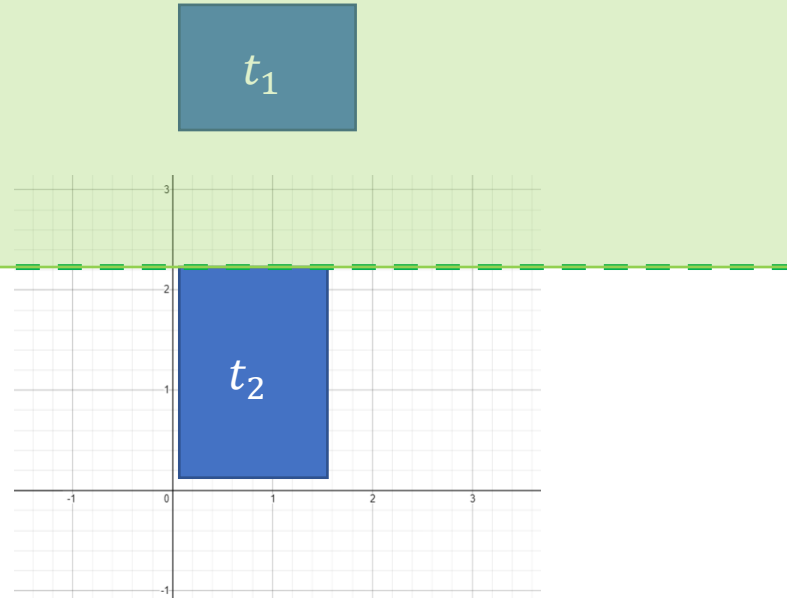
Intuition – 4 possible situations

$$y_1 + h_1 \leq y_2$$



Intuition – 4 possible situations

$$y_2 + h_2 \leq y_1$$



Non-overlapping constraints

In fact for the 4 constraints, only one need hold to allow for all locations that don't overlap!

Either

$$x_1 + w_1 \leq x_2$$

Or

$$x_2 + w_2 \leq x_1$$

Or

$$y_1 + h_1 \leq y_2$$

Or

$$y_2 + h_2 \leq y_1$$

Binary Variables

Introduce binary variables for the two tables

$$x_{12}, y_{12} \in \{0,1\}$$

We can use these to only consider one constraint, let M be a sufficiently large number

x_{12}	y_{12}	$M(x_{12} + y_{12})$	$M(1 - x_{12} + y_{12})$	$M(1 + x_{12} - y_{12})$	$M(2 - x_{12} - y_{12})$
0	0	0	M	M	$2M$
1	0	M	0	$2M$	M
0	1	M	$2M$	0	M
1	1	$2M$	M	M	0

Non-overlapping constraints

Our 4 constraints for table 1 and 2 are now

$$x_1 + w_1 \leq x_2 + M(x_{12} + y_{12})$$

$$x_2 + w_2 \leq x_1 + M(1 - x_{12} + y_{12})$$

$$y_1 + h_1 \leq y_2 + M(1 + x_{12} - y_{12})$$

$$y_2 + h_2 \leq y_1 + M(2 - x_{12} - y_{12})$$

x_{12}	y_{12}	C1	C2	C3	C4
0	0	Active	Satisfied	Satisfied	Satisfied
1	0	Satisfied	Active	Satisfied	Satisfied
0	1	Satisfied	Satisfied	Active	Satisfied
1	1	Satisfied	Satisfied	Satisfied	Active

Non-overlapping constraints

How many non overlapping constraints?

Test each pair. Given n items (tables)

$$4 \cdot \binom{n}{2}$$

This is a combinatorial problem!

Objective

For all i (every table)

$$x_i + w_i \leq A^*$$

$$y_i + h_i \leq A^*$$

Then

minimise A^*

This is known as a minimax problem. That is, we wish to minimise the upper A^* bound of the constraints

MILP Formulation

minimise A^*

subject to:

For all i (every table)

$$x_i + w_i \leq A^*$$

$$y_i + h_i \leq A^*$$

MinMax Constraint

For all $i, j, i < j$ (all pairs, don't check twice)

$$x_i + w_i \leq x_j + M(x_{ij} + y_{ij})$$

$$x_j + w_j \leq x_i + M(1 - x_{ij} + y_{ij})$$

$$y_i + h_i \leq y_j + M(1 + x_{ij} - y_{ij})$$

$$y_j + h_j \leq y_i + M(2 - x_{ij} - y_{ij})$$

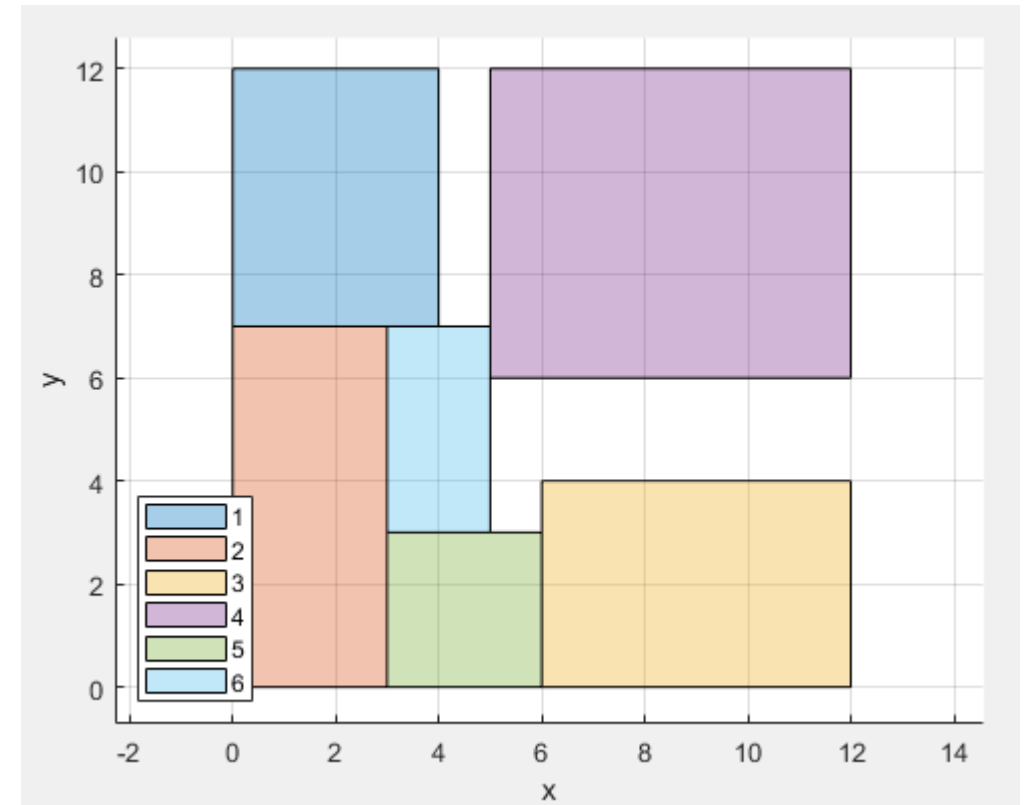
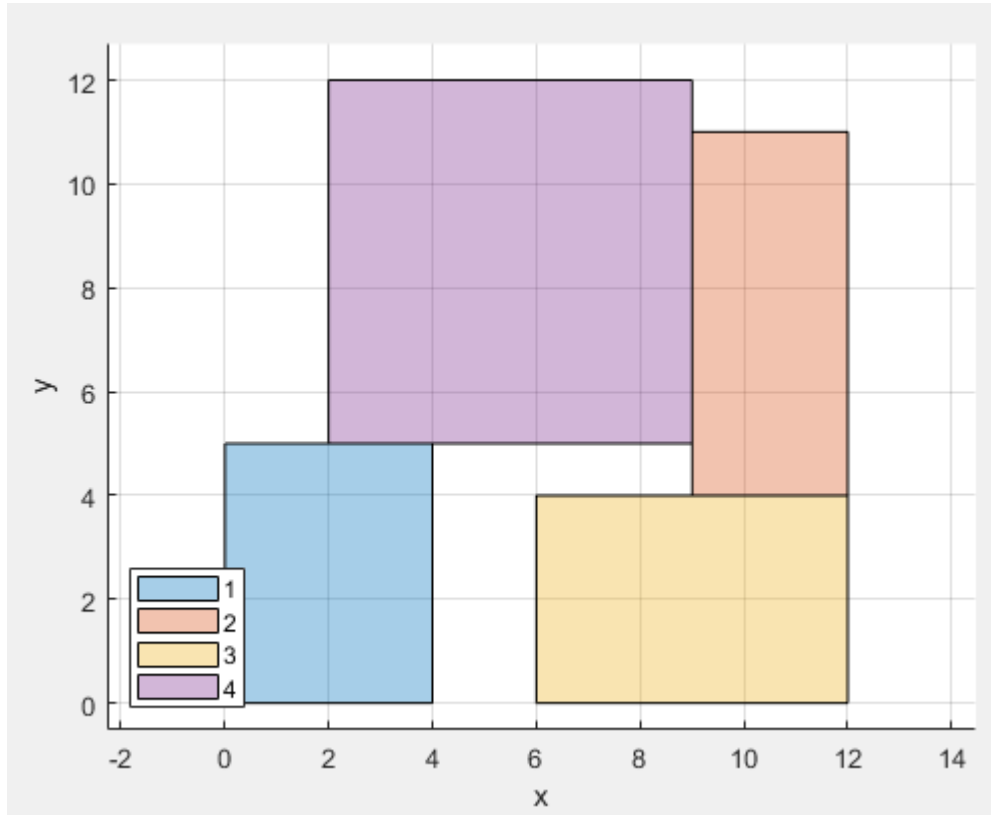
Non overlapping constraints

$$x_{ij}, y_{ij} \in \{0,1\}$$

$$x_i, y_i \geq 0$$

Variable constraints

Solution



Here is our solution, $A^* = 12$ for both 4 and 6 items!

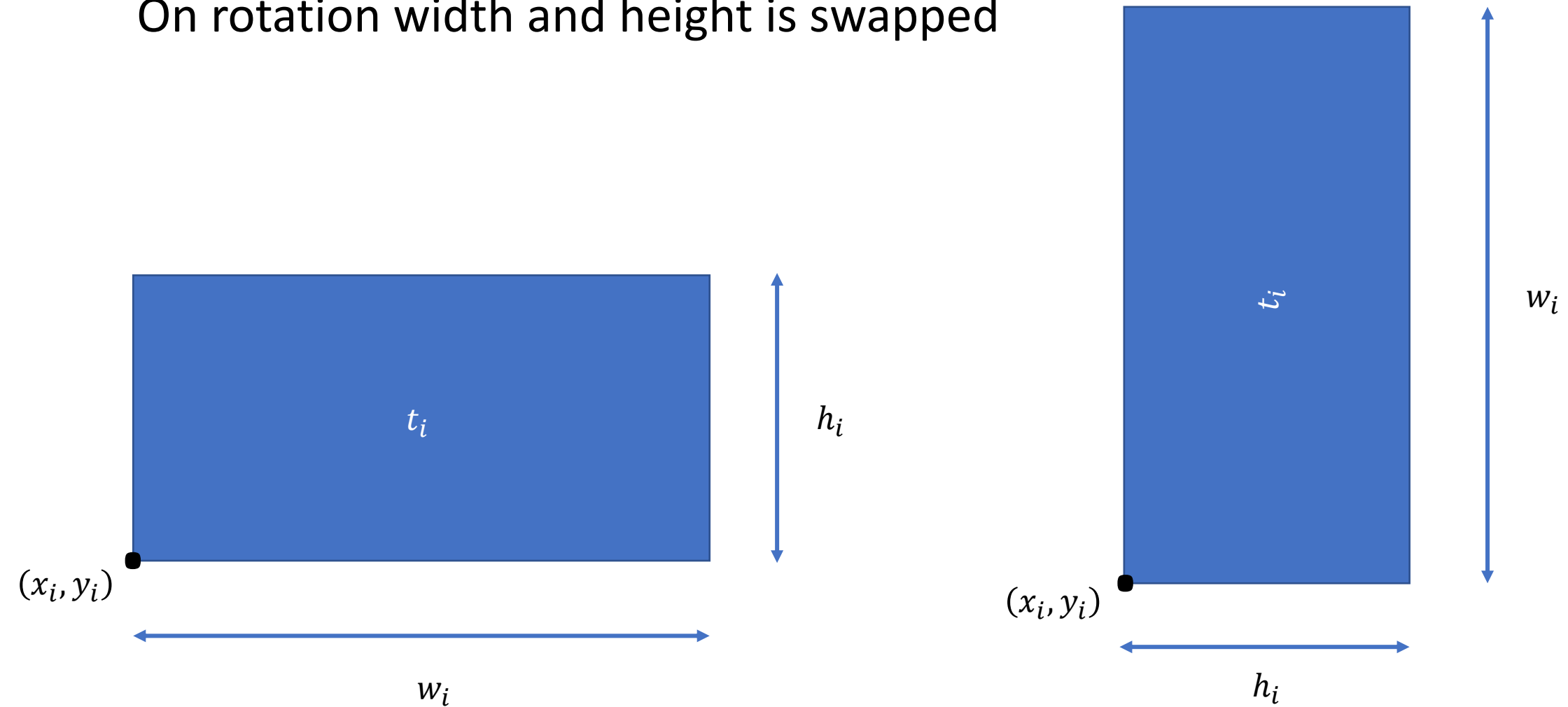
Our solutions are optimal

Can it be improved? What is the issue?

Adding Rotation

Floor Planning

On rotation width and height is swapped



Amending the constraints

Introduce binary variables r_i . $r_i = 0$ (not rotated), $r_i = 1$ (rotated).

Minimax constraints

For all i (every table)

$$x_i + (1 - r_i)w_i + r_i h_i \leq A^*$$

$$y_i + (1 - r_i)h_i + r_i w_i \leq A^*$$

Non overlapping constraints

For all $i, j, i < j$ (all pairs, don't check twice)

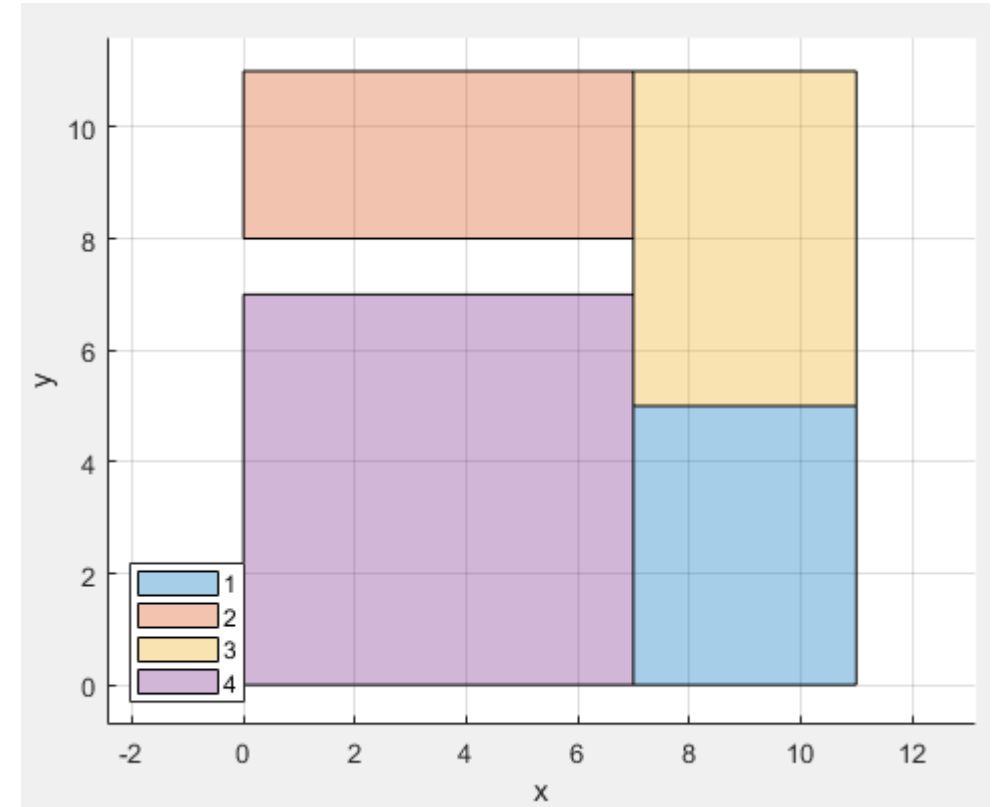
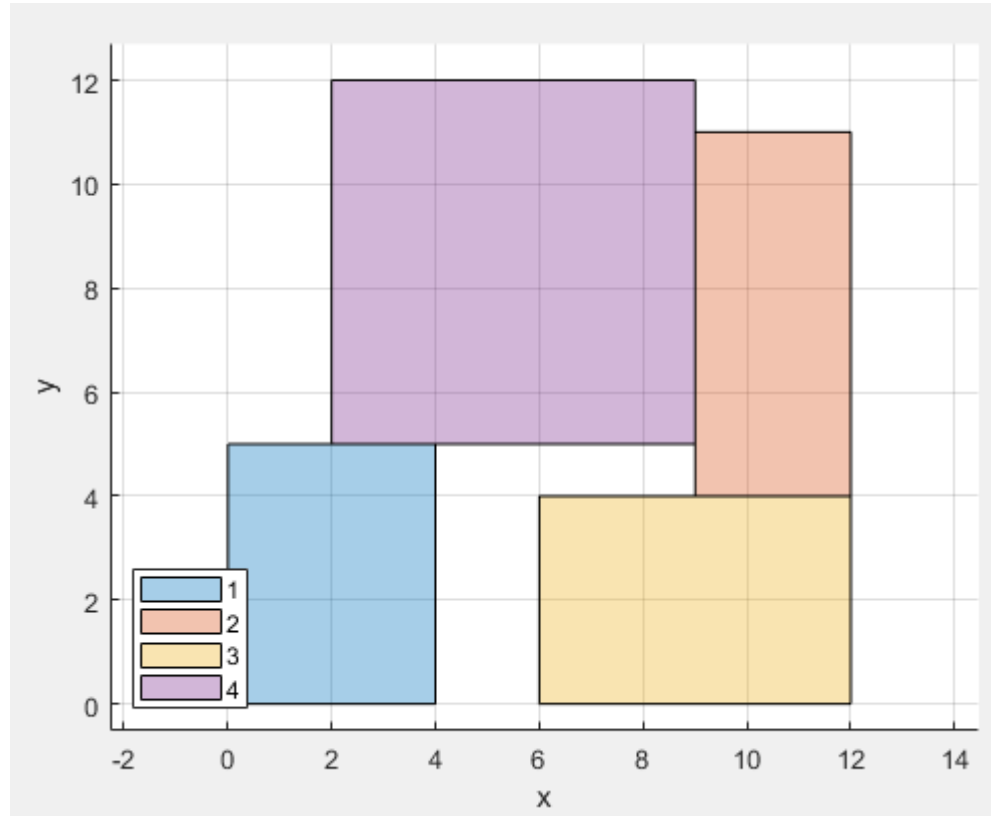
$$x_i + (1 - r_i)w_i + r_i h_i \leq x_j + M(x_{ij} + y_{ij})$$

$$x_j + (1 - r_j)w_j + r_j h_j \leq x_i + M(1 - x_{ij} + y_{ij})$$

$$y_i + (1 - r_i)h_i + r_i w_i \leq y_j + M(1 + x_{ij} - y_{ij})$$

$$y_j + (1 - r_j)h_j + r_j w_j \leq y_i + M(2 - x_{ij} - y_{ij})$$

Solution



Here is our solution, $A^* = 11$. Improvement!

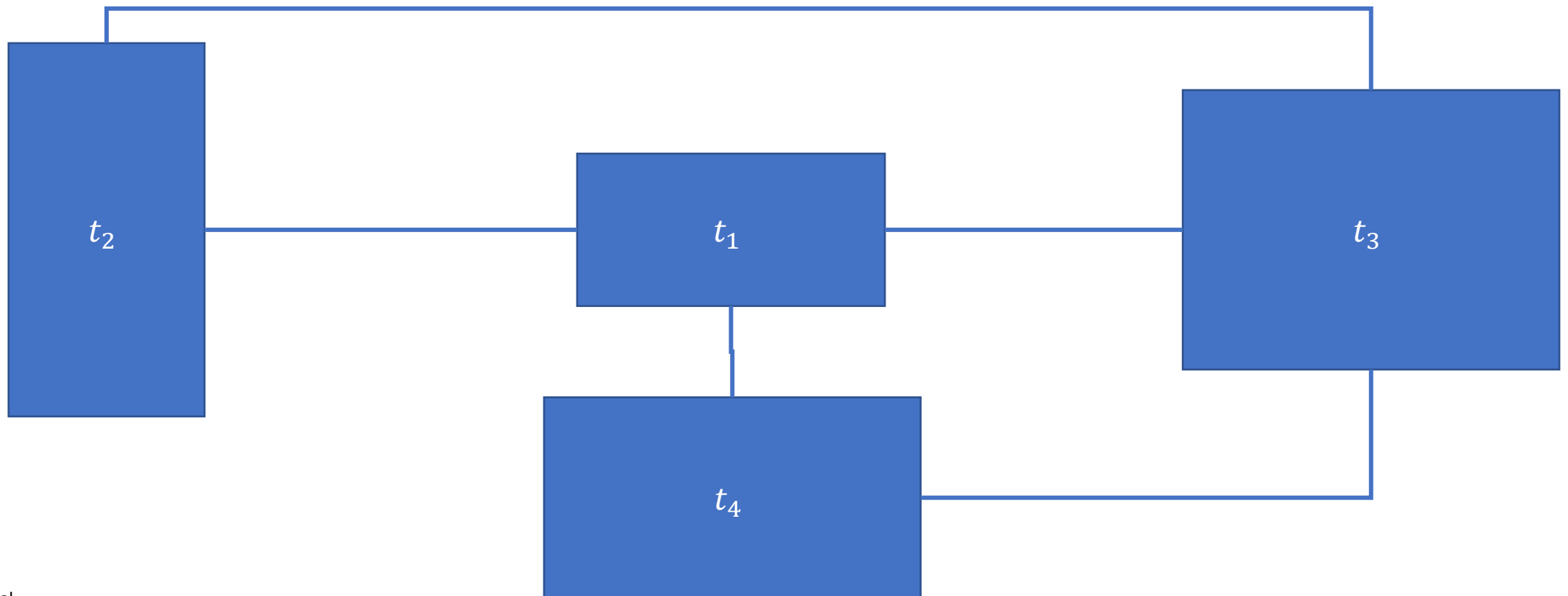
Can it be improved? This is dependent on your desired objective!

Connected 2 Dimensional Packing

Minimisation of the rectilinear distance between connected items

Clustering of components

Perhaps you have connected components that need to be close to each other and you have a fixed width/height for your floor space.



Amending the constraints

For a fixed width and height floor.

Minimax constraints

For all i (every table)

$$x_i + (1 - r_i)w_i + r_i h_i \leq W$$

$$y_i + (1 - r_i)h_i + r_i w_i \leq H$$

Note we could drop these if there was no constraints on the being within a boundary or make W, H sufficiently large!

Non overlapping constraints

For all $i, j, i < j$ (all pairs, don't check twice)

$$x_i + (1 - r_i)w_i + r_i h_i \leq x_j + M(x_{ij} + y_{ij})$$

$$x_j + (1 - r_j)w_j + r_j h_j \leq x_i + M(1 - x_{ij} + y_{ij})$$

$$y_i + (1 - r_i)h_i + r_i w_i \leq y_j + M(1 + x_{ij} - y_{ij})$$

$$y_j + (1 - r_j)h_j + r_j w_j \leq y_i + M(2 - x_{ij} - y_{ij})$$

Distance between items (from centre points)

For all connected i, j

$$\text{Let } Dx_{ij} = \left[x_i + (1 - r_i) \frac{w_i}{2} + r_i \frac{h_i}{2} \right] - \left[x_j + (1 - r_j) \frac{w_j}{2} + r_j \frac{h_j}{2} \right]$$

We need absolute distance, therefore we force $R_{ij} = \max\{Dx_{ij}, -Dx_{ij}\}$

$$\begin{aligned} Dx_{ij} &\leq R_{ij} \\ -Dx_{ij} &\leq R_{ij} \end{aligned}$$

$$\text{Let } Dy_{ij} = \left[y_i + (1 - r_i) \frac{h_i}{2} + r_i \frac{w_i}{2} \right] - \left[y_j + (1 - r_j) \frac{h_j}{2} + r_j \frac{w_j}{2} \right]$$

We need absolute distance, therefore we force $B_{ij} = \max\{Dy_{ij}, -Dy_{ij}\}$

$$\begin{aligned} Dy_{ij} &\leq B_{ij} \\ -Dy_{ij} &\leq B_{ij} \end{aligned}$$

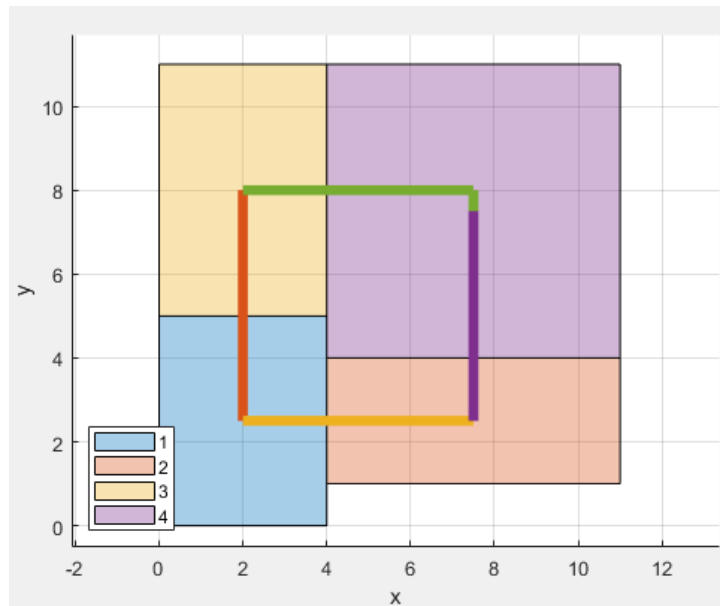
Objective Function

Minimise Rectilinear Distance

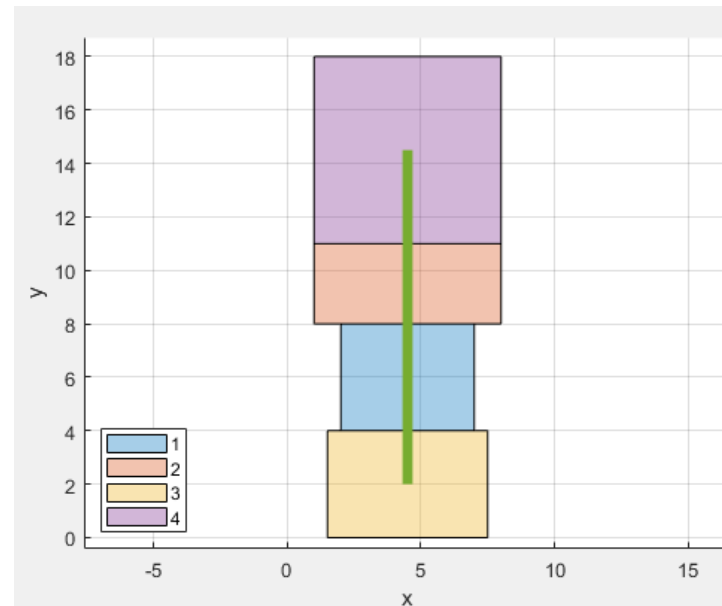
For all connected i, j

$$\text{Min } \sum_i \sum_j (R_{ij} + B_{ij})$$

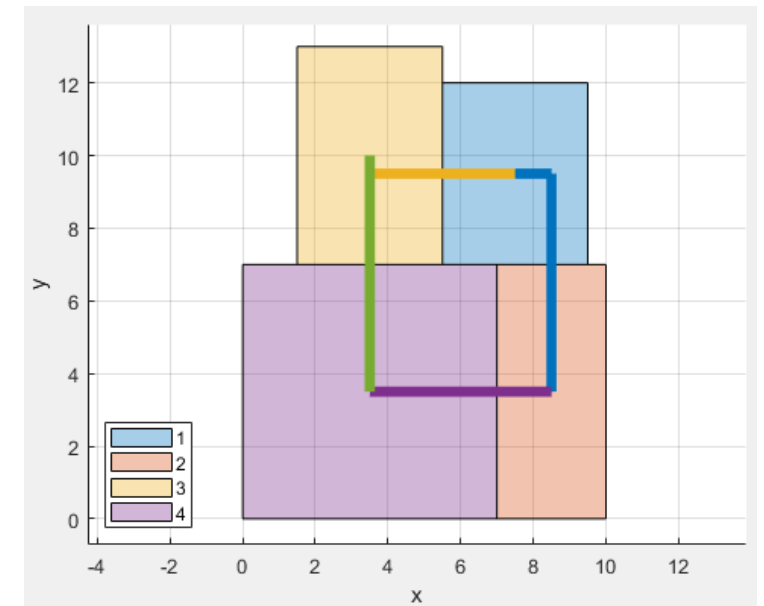
Results for different floor spaces



$W = 20,$ $H = 20$

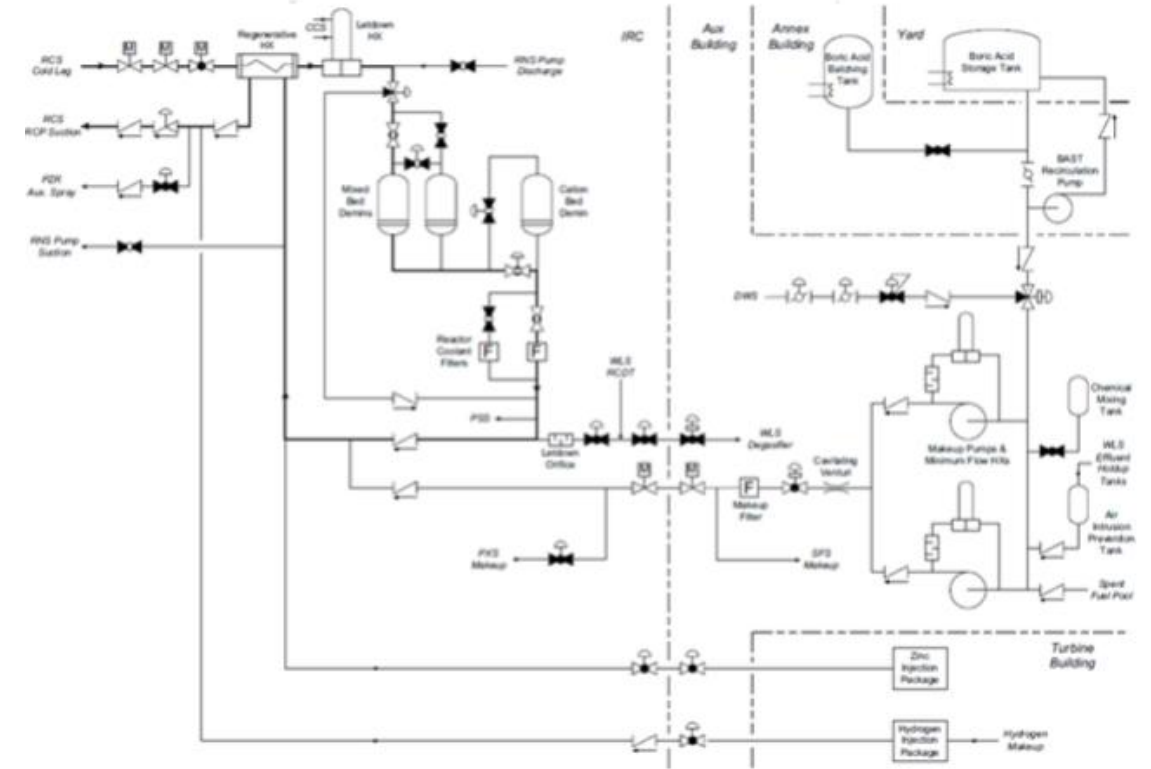


$W = 8,$ $H = 20$

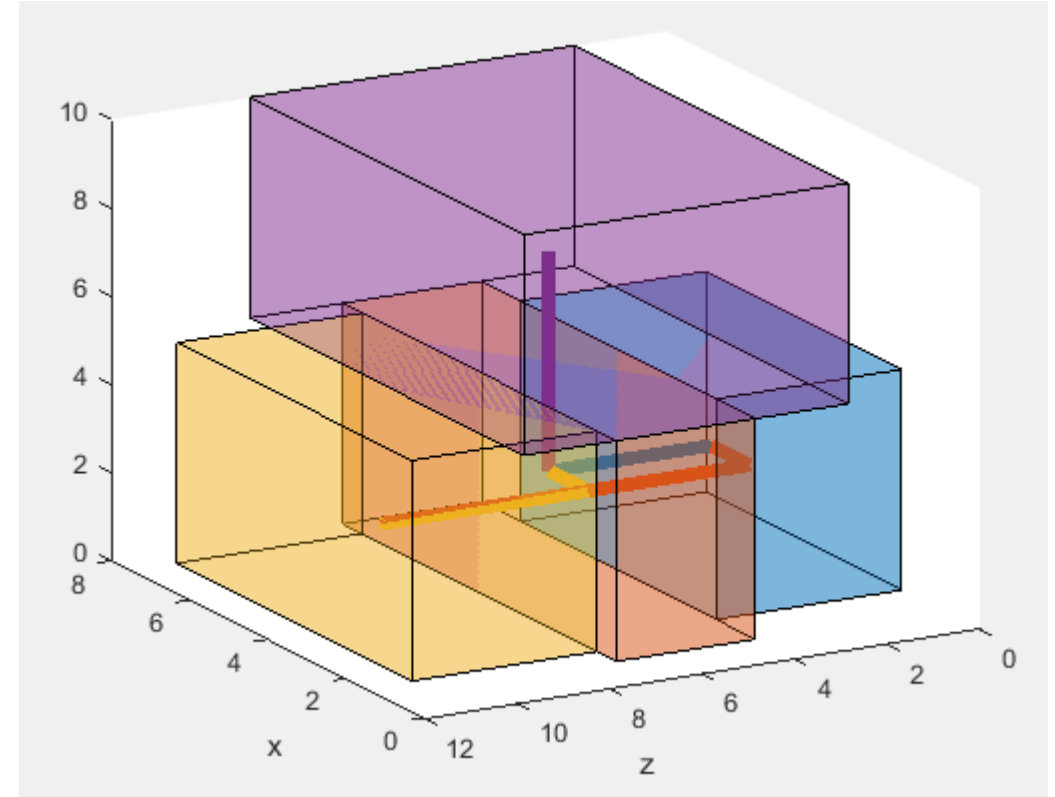
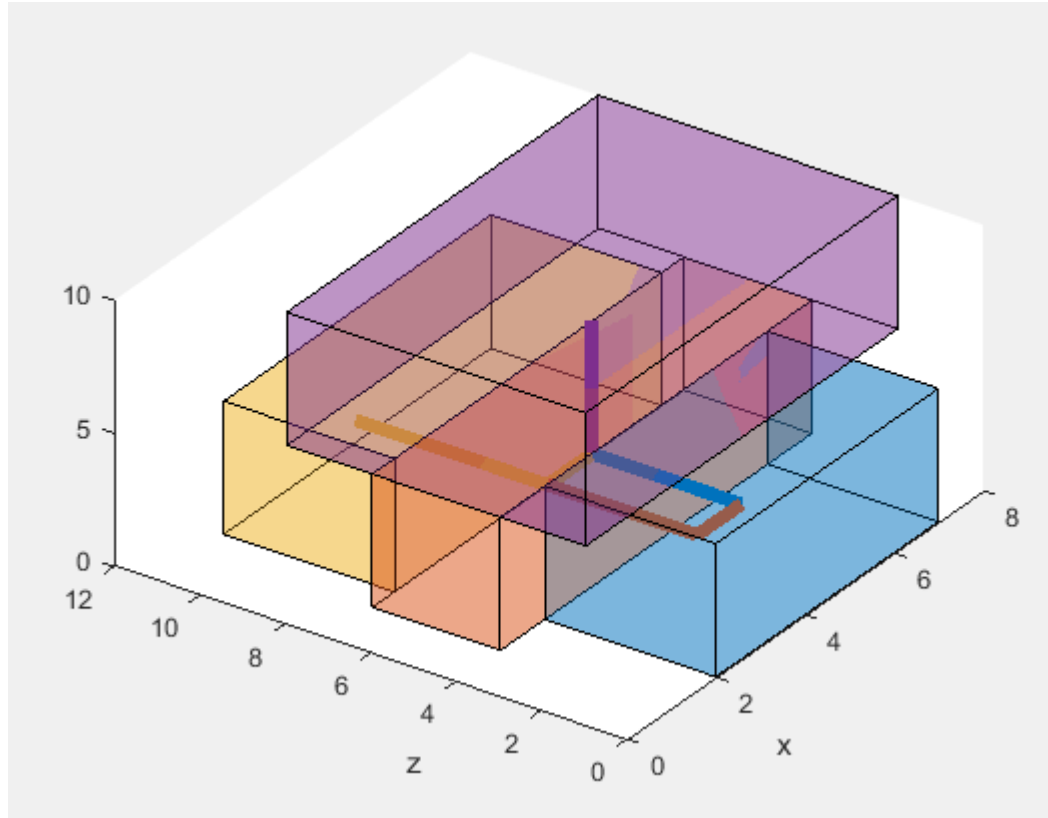


$W = 10,$ $H = 15$

Problem



Example Solution



This is a simplified plot, the boxes contain enough space to include the piping.
It also requires structures to support items off the floor!

Returning to our simple packing problem

minimise A^*

subject to:

For all i (every table)

$$x_i + (1 - r_i)w_i + r_i h_i \leq A^*$$

$$y_i + (1 - r_i)h_i + r_i w_i \leq A^*$$

MinMax Constraint

For all $i, j, i < j$ (all pairs, don't check twice)

$$x_i + (1 - r_i)w_i + r_i h_i \leq x_j + M(x_{ij} + y_{ij})$$

$$x_j + (1 - r_j)w_j + r_j h_j \leq x_i + M(1 - x_{ij} + y_{ij})$$

$$y_i + (1 - r_i)h_i + r_i w_i \leq y_j + M(1 + x_{ij} - y_{ij})$$

$$y_j + (1 - r_j)h_j + r_j w_j \leq y_i + M(2 - x_{ij} - y_{ij})$$

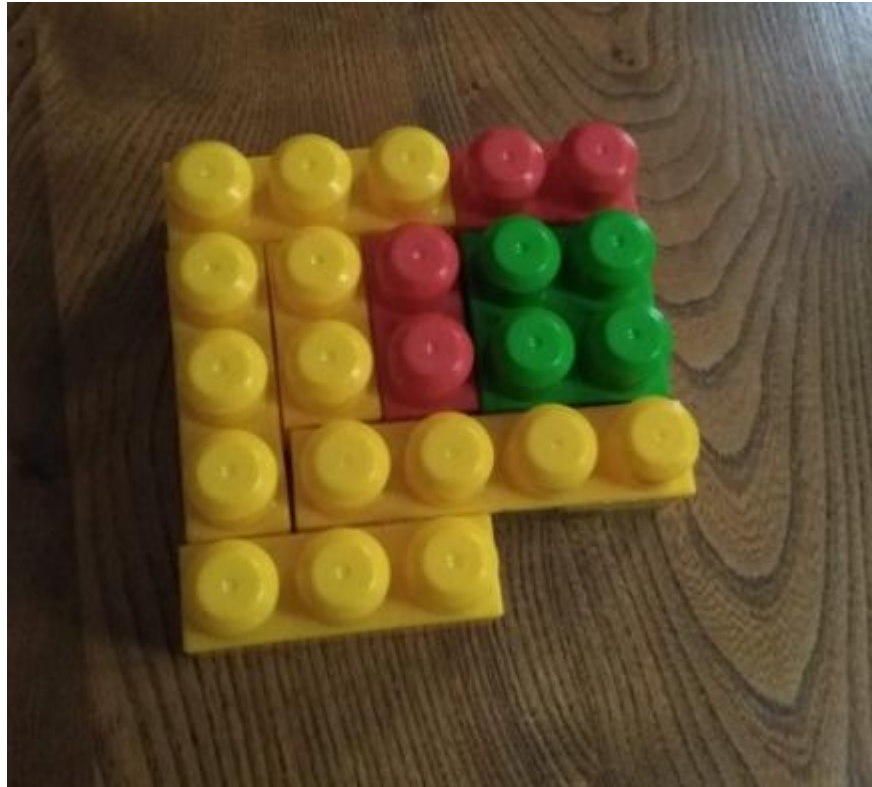
Non overlapping constraints

$$x_{ij}, y_{ij} \in \{0,1\}$$

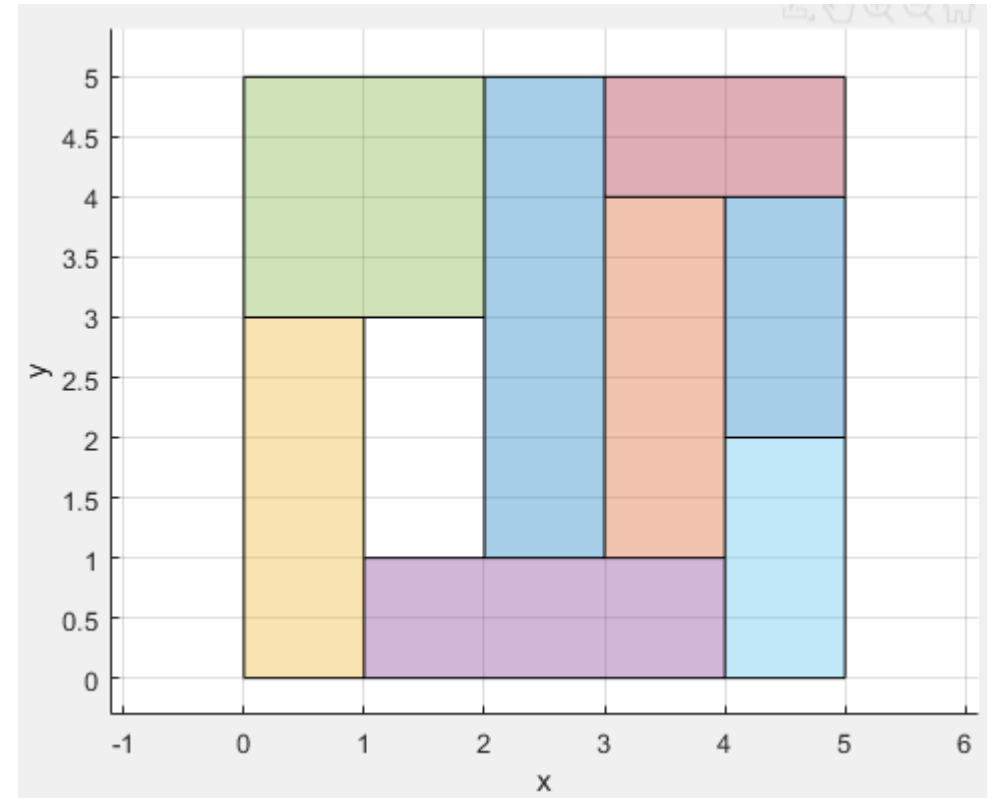
$$x_i, y_i \geq 0$$

Variable constraints

Mums Know Best!



- Approximately 20 seconds
- Approximately 4 solutions explored



- Approximately 10 minutes!
- 4,936,427 solutions explored
- Note that the optimal was actually found in 0.6 seconds, but the program doesn't know it's optimal

Why so inefficient?

Binary variables.

x_{ij}, y_{ij} exist for every pair of items

$$2 \cdot \binom{n}{2}$$

Each can have a value of $\{0,1\}$

$$2^{2 \cdot \binom{n}{2}}$$

For 7 items that is...

$$2^{2 \cdot \binom{7}{2}} \approx 4.4 \times 10^{12}$$

For 10 items to explore every solution every millisecond this would take you

39228790500000000 years!

Conclusion

How does Santa pack his sleigh?

By Magic!

Or NP-Hard Optimisation!

